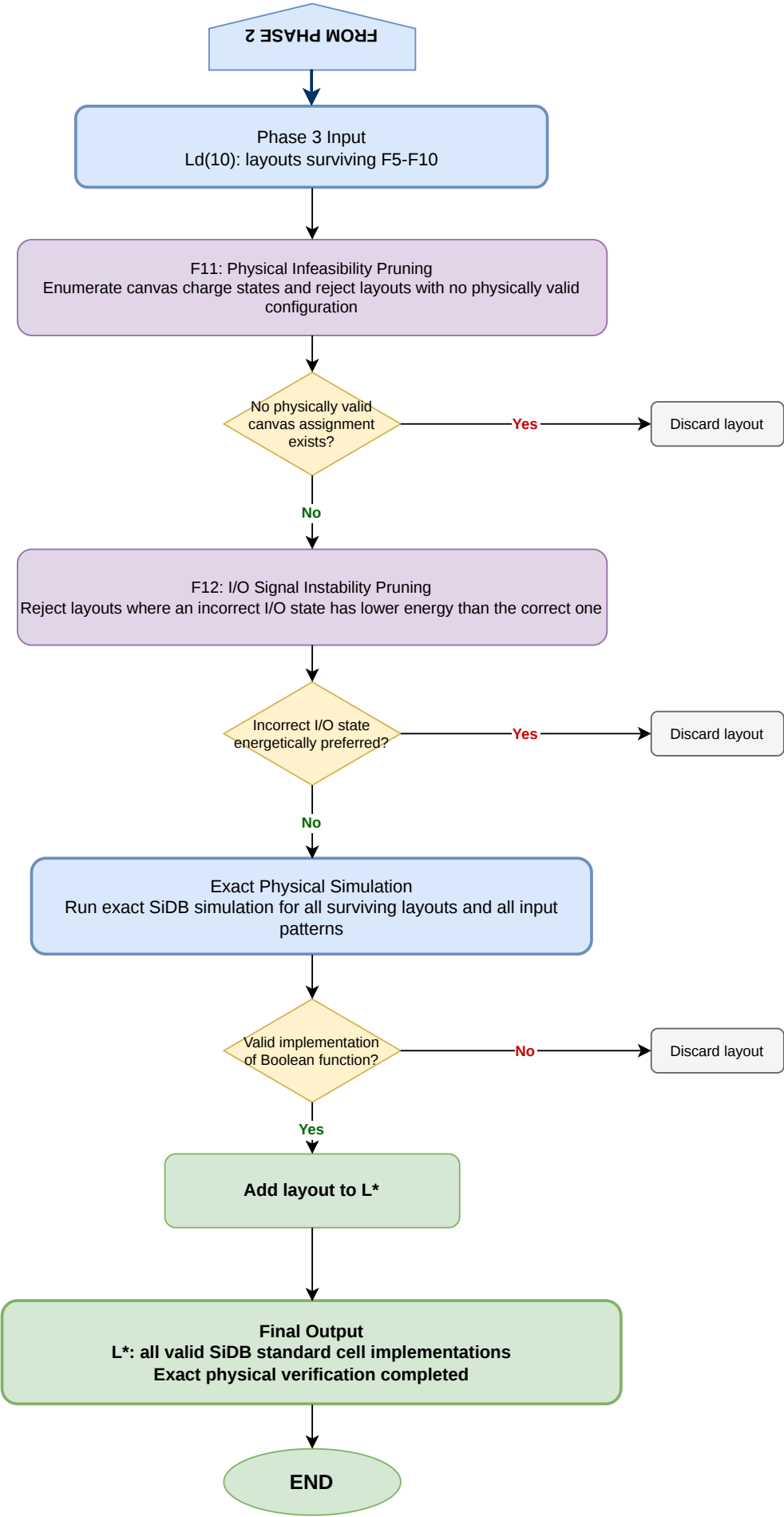


Phase 3: Enumerative Physical Pruning (F11-F12) + Exact Physical Simulation
QuickCell upgraded from 3 filters to 12 filters



Phase 3 Summary

Filters: F11 and F12
Then: Exact Physical Simulation
Role: Remove layouts with no physically valid configuration or unstable correct I/O state, then exactly verify remaining layouts
Output set: L*

This phase preserves the original QuickCell philosophy while extending the pruning pipeline to 12 filters.